Week #8: 19 Oct- 23 Oct

Subject/:الموضوع Chemistry

Unit Title: عنوان الوحدة Unit 4: Oxidation and reduction

Grade الصف: 10

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: Balancing redox reaction in acidic and basic medium	I can balance redox (oxidation–reduction) reactions in acidic and basic solutions by adding water molecule to balance H+ and OH- ions.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 4.2 Ensure access to tools and assistive technologies.	McGraw Hill Module 18, PPT; Chemistry Notebook; Teams; Toddle
Lesson 2 Practice, Revise, Revisit	I can analyze and apply my chemistry knowledge to solve exam-style questions, identify gaps in my understanding, and improve my work using feedback.		
Homework الواجبات	Review **oxidation, reduction, electron transfer, and oxidation-number rules**. Complete unfinished notebook/class practice.		
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Hello Parents, Please encourage your child to review and practice Science regularly each week using the resources available in their class Microsoft Teams - class material folder which is organized by week. The presentations, readings, videos, practice activities, etc. posted in Teams are the primary learning materials and should be used throughout the unit to support understanding and achievement. Science study guides will provide basic *assessment information, topics, and expectations*; however, they are intended as a guide only and will not contain detailed study materials, these are found in the class teams. Effective preparation for graded formatives and summative requires students to actively engage with the learning materials provided in their Toddle class drive. Thank you for partnering with us to support your child's learning and success. Ms. Stacey (Science Coordinator)		

#الأسبوع Week 30

Technology Integration دمج التكنولوجيا	Use **digital reaction models/PPT animations** to visualize electron transfer and changes in oxidation numbers during redox reactions. Students may use digital periodic-table resources to support oxidation-number determination.
Cross-curricular links الروابط بين المواد الدراسية	**Physics: movement and transfer of electrons and electrical charge.** **Mathematics: applying positive and negative numbers and algebraic reasoning to determine unknown oxidation numbers.** **Environmental Science: oxidation processes such as corrosion and the role of redox chemistry in batteries and environmental systems.**
External Assessment Integration دمج التقييمات الخارجية	MAP , CAT 4
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Thinking: analyze electron transfer and oxidation-number changes.** **Communication: explain redox reactions using correct scientific vocabulary.** **Self-management: follow a systematic sequence when assigning oxidation numbers and analyzing reactions.** **Collaboration: compare solutions and justify reasoning during differentiated problem-solving activities.**
Standards المعايير	**HS-PS1-2: Construct and revise an explanation for the outcome of a simple chemical reaction based on the outermost electron states of atoms, trends**
Inquiry/ TOK Question البحث والاستقصاء أسئلةTOK	**How can we use evidence that we cannot directly see, such as electron transfer and changes in oxidation numbers, to explain what happens during a redox reaction?**

Success Criteria معايير النجاح (Measurable)	لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Lesson 1-** • **STUDENT MUST identify which substances are oxidised and reduced and assign correct oxidation numbers in a redox reaction.** • **STUDENT SHOULD balance simple redox reactions by adjusting coefficients so the number of electrons lost equals the number gained.** • **STUDENT COULD balance more complex redox reactions (including ionic equations) and explain the transfer of electrons using correct scientific terminology** **Lesson 2 —** • MUST I can recall and apply key chemistry concepts to answer exam-style questions. • SHOULD I can analyze exam-style questions, select the correct scientific knowledge, and justify my answers using appropriate chemistry terminology. • COULD I can evaluate my answers, identify my learning gaps, and independently improve or complete formative tasks on Toddle using feedback and lesson resources.
National Identity Domain مجالات الربط بالهوية الوطنية	Conservation الحفاظ ع للبيئة Global Understandingه التفاهم العال Heritage ال رتاث **Conservation الحفاظ على البيئة | Global Understanding التفاهم العالمي** **UAE Values & Sustainable Development** Connect redox chemistry to **battery technology, corrosion prevention, and responsible battery disposal in the UAE**. Students discuss how understanding oxidation and reduction can support **safer energy use, protection of infrastructure, and environmental responsibility**. This also connects nicely with the **responsible battery disposal and environmental responsibility** point already included in the previous planner format